

Ian Foy

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Engineering, Electrical Engineering

Expected Graduation: December 2025

GPA: 3.5 / 4.0

Relevant Coursework: Advanced Embedded Systems, Digital Logic Design, Electrical Engineering Systems Design, Embedded Control Systems, & Programming & Data Structures

WORK EXPERIENCE

General Motors

Warren, MI

Assistant Design Release Engineer

June 2025- August 2025

- Resolved incorrect front hood latch service procedures by leading weekly meetings & verifying electrical diagrams to align instructions with actual part specs.
- Corrected requirement documents within the component technical specification for outside door handles to ensure traceability, prevent supplier confusion, & reduce risk of future recalls.

General Motors

Warren, MI

Assistant Vehicle Systems Engineer

May 2024- August 2024

- Developed Python code to map option statements to electronic control units, enabling vehicle systems engineers to efficiently access, & interpret data.
- Initiated improvements to the owner's manual review website, accounting for differences in vehicle electrical architectures.

General Motors

Warren, MI/Milford, MI

Hardware Engineering Intern

May 2023- August 2023

- Modified production line requirements in IBM Rational DOORs to address safety & power consumption issues, improving vehicle efficiency.
- Created a feature model to automate 4,000+ calibration values across 12 vehicle platforms, increasing configuration accuracy.

PROJECTS

Baja SAE- CAN-Based Data Collection System

- Built a CAN-based data acquisition system enabling real-time monitoring of vehicle dynamics, GPS position, & suspension travel to support performance tuning & race diagnostics.
- Implemented embedded software to interface with linear potentiometers & accelerometers for accurate measurement of suspension & body dynamics.
- Designed & tested PCBs for the suspension measurement & vehicle dynamics modules using Altium.

Computer Vision-Controlled Guitar Effects System

- Developed a Raspberry Pi-based computer vision system using OpenCV to detect guitar tilt gestures in real time & trigger audio effects via UART communication with an STM32-controlled amplifier.
- Designed & built an STM32-controlled guitar effects system, implementing analog switching to toggle between clean, delay, & distortion signal paths with preamp, amplifier, & PT2399 delay modules.

Music Visualizer

- Led development of a music visualizer using DSP, autonomous robotics, an LED equalizer, & an LCD.
- Wrote embedded software for Zumo robot movement using QTR-8RC sensor, H-Bridge motor driver, & XBee communication, integrating hardware via GPIO, UART.

SKILLS

- **Programming Languages:** Python, C++, C, Verilog, ARM assembly, Matlab, Bash Scripting
- **Software & Tools:** Logic Analyzer, Oscilloscope, NanoVNA, IBM Rational DOORs, Vehicle Spy, Git
- **Technical Skills:** STM32 & NXP microcontrollers, I2C, UART, SPI, GPIO, ADC, PCB design, FreeRTOS, Linux, API design, FPGAs, Bare-metal Programming, rapid prototyping, electrical systems diagnostics
- **Other Skills:** Excellent project management, communication, problem-solving, & collaboration skills